

# ZeroTrustBlock

## Z12 End-to-End Evaluation

Local cryptographic and application validation, reproducible evaluation assets, and live Fabric readiness assessment

| Decision               | Scope                                                                                | Evaluation revision                      |
|------------------------|--------------------------------------------------------------------------------------|------------------------------------------|
| PASS WITH OBSERVATIONS | Local ZKP, Gateway, and chaincode checks pass; live Fabric measurements unavailable. | 0fe94ecc7eeb5aa03d4a74b7b770690263bae7ed |

**Evaluation date:** 04 October 2026 (IST)

**Repository:** ZeroTrustBlock thesis repository

**Report purpose:** Record measured local behavior and distinguish it from unmeasured live-network behavior. No live Fabric deployment or lifecycle mutation was attempted.

### Decision rationale

All available local ZKP, Gateway, and chaincode test suites passed. Deterministic adversarial cases and authority-key validation passed, and benchmark samples were collected. The Docker client could not access the Docker daemon, so Fabric-backed end-to-end execution, ledger commit latency, concurrency behavior on peers, and live storage inspection could not be established. The decision therefore has an explicit live-network observation.

# 1. Scope and environment

Z12 validates the existing HealthClaimHardened flow through local cryptographic checks and Gateway/chaincode unit tests; prepares gated Fabric integration checks; measures local proof and Gateway processing; and records privacy/storage sizes from a deterministic serializer fixture. No circuit redesign or changes to Core, Gateway architecture, Fabric architecture, IPFS, Docker, networking, or Federated Learning were undertaken. The only chaincode implementation change is authority-key validation hardening plus read-only query methods needed by the gated live harness.

| Item                               | Observed value                                                                                                      |
|------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| OS / kernel                        | Ubuntu 24.04.5; Linux 6.17.0-1032-oem; x86_64                                                                       |
| CPU / RAM                          | AMD Ryzen 7 PRO 8840HS; 8 cores / 16 threads; 27 GiB RAM                                                            |
| Go                                 | Host Go 1.22.2; module toolchain Go 1.25.8                                                                          |
| Crypto dependencies                | gnark v0.16.3; gnark-crypto v0.21.0                                                                                 |
| Fabric tooling                     | CLI peer 2.4.9; Docker client 29.1.3; standalone docker-compose 1.29.2                                              |
| Docker daemon                      | Unavailable to this process: permission denied at /var/run/docker.sock                                              |
| Declared topology                  | HospitalMSP and InsurerMSP; 4 peers; 3 Raft (etcdraft) orderers; healthchannel; health chaincode                    |
| Declared policy / TLS              | AND(HospitalMSP.member, InsurerMSP.member); TLS enabled in deployment/profile                                       |
| Live network identity and sequence | Unverified. Repository declares Fabric 2.4.9, CA 1.5.7, chaincode v1.0 seq 1; no claim about running network state. |

The scripts run a read-only preflight and deliberately do not start, reset, deploy, or upgrade a network. This avoids accidentally applying the repository deployment script against a possibly existing channel.

# 2. Local test and constraint results

| Suite                         | Result                                                                                                                                                                                                            |
|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| HealthClaimHardened           | PASS. Deterministic relation test reports 22,021 constraints, 13 public variables, and 31 secret variables.                                                                                                       |
| Gateway packages              | PASS. Includes adversarial proof/input mutation, replay/nullifier, trusted-anchor, and fail-closed checks.                                                                                                        |
| Chaincode                     | PASS. Includes claim/challenge/nullifier handling, authorization, privacy serialization, and authority-key validation tests.                                                                                      |
| Gated live test compilation   | PASS. Built with z12integration tag; live tests were not executed.                                                                                                                                                |
| Circuit adversarial witnesses | PASS for the tested negative cases: range overflow, invalid signatures/keys, wrong anchors/versions/context, non-Boolean selector, invalid roots, challenge/recipient/domain mismatch, and unsupported diagnosis. |

Constraint count and variable counts are from the local HealthClaimHardened test output. They are not a count for Fabric or an end-to-end transaction. Full raw logs are in output/z12/.

### 3. Local performance

Benchmarks used 10 independent runs with 10 timed operations per run and GOMAXPROCS=16. P95 is nearest-rank and equals the maximum for N=10. P99 is intentionally omitted because the sample count is too small to support it. These are local CPU measurements; none include endorsement, ordering, network transport, or commit.

| Operation             | Mean      | Median    | Min       | Max / P95 | Sample SD |
|-----------------------|-----------|-----------|-----------|-----------|-----------|
| Proof generation      | 70.702 ms | 70.191 ms | 67.287 ms | 76.695 ms | 2.777 ms  |
| Proof verification    | 0.675 ms  | 0.676 ms  | 0.630 ms  | 0.711 ms  | 0.023 ms  |
| Gateway witness build | 0.0026 ms | 0.0022 ms | 0.0011 ms | 0.0056 ms | 0.0016 ms |
| Gateway prove         | 75.712 ms | 75.001 ms | 74.523 ms | 78.958 ms | 1.548 ms  |
| Gateway local verify  | 0.987 ms  | 0.980 ms  | 0.900 ms  | 1.096 ms  | 0.066 ms  |
| Gateway local total   | 77.184 ms | 76.395 ms | 75.956 ms | 80.501 ms | 1.590 ms  |

#### Z9 baseline comparison

| Measure             | Z9 baseline | Z12 local                     | Difference                 |
|---------------------|-------------|-------------------------------|----------------------------|
| Proof generation    | 73.575 ms   | 70.702 ms                     | -2.873 ms (-3.9%)          |
| Proof verification  | 0.693 ms    | 0.675 ms                      | -0.018 ms (-2.7%)          |
| Full local pipeline | 76.320 ms   | 77.184 ms Gateway-local total | +0.864 ms descriptive only |

The samples and harnesses are small/different enough that these deltas are descriptive, not evidence of a performance improvement or regression. Derived sequential Gateway-local throughput is approximately 12.96 operations/s (1 / mean local total); it is not distributed throughput and excludes Fabric. No live Fabric latency, throughput, endorsement, orderer, or commit values are reported.

### 4. Fabric end-to-end status

| Requested measurement                          | Z12 result                                                                     |
|------------------------------------------------|--------------------------------------------------------------------------------|
| Positive claim acceptance and ledger queries   | NOT MEASURED: live Fabric inaccessible. Gated harness is present and compiles. |
| Invalid proof rejection at submission          | NOT MEASURED live. Local Gateway/chaincode negative tests pass.                |
| Sequential claims and duplicate-nullifier race | NOT MEASURED live. Gated test is prepared; no peer/orderer race result.        |
| Endorsement / submit / commit latency          | NOT MEASURED.                                                                  |
| Live proof size / state growth                 | NOT MEASURED. Storage figures below are fixture serialization only.            |

## 5. Security and privacy observations

The circuit and surrounding components enforce different parts of the protocol. The circuit proves the private witness relation under its public inputs. Gateway checks local proof validity and trusted local anchors before submission. Chaincode checks caller role, anchored state, proof metadata/format, and duplicate challenge/nullifier state; it does not verify the ZK proof itself. Trust-anchor registration remains governed by authorized HospitalMSP administration and ledger governance.

| Area                   | Local evidence / boundary                                                                                                                                                                                                                                                                                                                                                |
|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Authority key          | Chaincode registration now rejects non-canonical coordinates, off-curve points, identity, and keys outside the prime-order subgroup. Tests reject (1,2), identity (0,1), and order-2 torsion (0,p-1), and accept a standard subgroup generator. Arithmetic is custom BN254 twisted-Edwards code; independent cryptographic review and live ledger test remain advisable. |
| Challenge / replay     | Unit tests show consumed challenges and used nullifiers are rejected. Fresh challenge generation and cross-ledger UsedNullifiers consistency remain external operational responsibilities. Tests do not prove distributed race behavior without a live network.                                                                                                          |
| Authority trust anchor | Circuit signature verification proves the signature against its public authority key. Chaincode registration/administrative authorization establishes which key is trusted; the external trust-anchor governance and key custody are not cryptographically established by the circuit.                                                                                   |
| Policy commitment      | Circuit proves the witness policy values match the public commitment used in the relation; Gateway/ledger anchor consistency is separately tested locally. Correct policy publication and governance remain external.                                                                                                                                                    |
| Context and bindings   | Local negative tests mutate recipient, domain, deployment, context commitment, challenge, root/policy versions, registry root, and nullifier. The tested circuit rejects these mismatches. ProtocolDomain is hardcoded as 202610041; replace with a stable, documented protocol identifier before production interoperability.                                           |
| Privacy boundary       | Diagnosis, lab value, patient secret, coverage ceiling, Merkle siblings/directions, and policy ID are absent from the modeled claim/audit records. Proof bytes and all 13 public inputs are channel-visible. Private data can still leak through client logs, endorsement inputs, application telemetry, or auxiliary systems unless those are controlled.               |

The deterministic proof relation, gateway/chaincode tests, and code inspection establish local properties only. They do not establish key custody, correct live channel configuration, organizational trust, deployment identity, ledger privacy configuration, or operational freshness.

## 6. Storage and evidence

| Serialized fixture field | Bytes / size                          |
|--------------------------|---------------------------------------|
| Proof bytes              | 164 bytes (placeholder fixture proof) |
| Proof hash hex           | 64 characters                         |
| Public statement JSON    | 425 bytes                             |
| Claim record JSON        | 1,050 bytes                           |
| Audit record JSON        | 301 bytes                             |

These are deterministic local serialization sizes from TestZ12StorageSerializationSizesAndPrivacy, not measurements of actual committed ledger values. The fixture explicitly confirms that private diagnosis/lab/secret/coverage/path/policy-ID fields are absent.

## 7. Reproduction

From the repository root, reproduce local validation and benchmarks with the scripts below. They store command output in `output/z12/`. Live scripts first run the read-only preflight and stop when the Docker daemon is inaccessible. For live execution, configure an already-running, correctly deployed Fabric network and the expected credentials/profile, then run the live scripts; do not assume the repository deploy script is safe to rerun on an existing channel.

| Purpose                                 | Command                                          |
|-----------------------------------------|--------------------------------------------------|
| Local tests + gated integration compile | <code>scripts/z12/run-local-validation.sh</code> |
| Local benchmarks                        | <code>scripts/z12/run-local-benchmarks.sh</code> |
| Inspect live readiness                  | <code>scripts/z12/network-preflight.sh</code>    |
| Live end-to-end claim                   | <code>scripts/z12/run-live-e2e.sh</code>         |
| Live negative / replay cases            | <code>scripts/z12/run-live-negative.sh</code>    |
| Live performance run                    | <code>scripts/z12/run-live-performance.sh</code> |
| Refresh raw result manifest             | <code>scripts/z12/collect-results.sh</code>      |

Raw logs, environment details, preflight output, benchmark summaries, storage result, and SHA-256 manifest are stored under `output/z12/`. The exact test and benchmark commands are recorded in the scripts and logs. A PDF hash is included in the final manifest.

## 8. Final decision

### PASS WITH OBSERVATIONS

Local implementation validation and tests pass, including 22,021 circuit constraints and the authority-key adversarial tests. Local proof and Gateway benchmarks completed. The report does not claim end-to-end Fabric success: the Docker daemon denied access, so live proof submission, ledger commit, concurrency/replay under endorsing peers, actual storage/privacy, and Fabric latency/throughput remain unverified. The next validation step is to run the provided live harness against a reachable, correctly configured Fabric network and retain its raw output.

### Artifacts

Repository changes are limited to Z12 evaluation support, authority-key registration hardening and query methods, Gateway phase timing, gated live integration tests, README and reproducibility scripts, raw result logs, and this report. See `README-Z12.md` and `output/z12/result-manifest.sha256`.